TryHackMe Writeup 8: Intro to LAN

Room: LAN topologies, switches and routers, subnetting, ARP, DHCP

Objective: Learn local area networks, network topologies, subnetting, ARP, and DHCP. These concepts explain how devices communicate within a local network and obtain network configuration

Step 1 - Learn LAN and network topologies

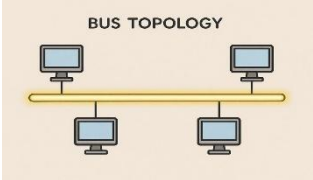
LAN stands for Local Area Network.

A LAN connects devices within a limited area, such as a home, office, or school.

Common topologies include:

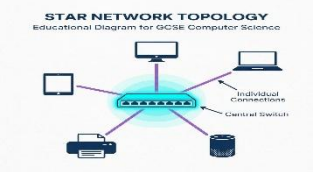
Bus topology

Devices share a common backbone. It can be inexpensive to set up, but backbone failures can disrupt communication.



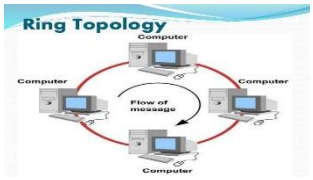
Star topology

Devices connect to a central switch or hub. It is common in modern LANs, but the central device is an important dependency.

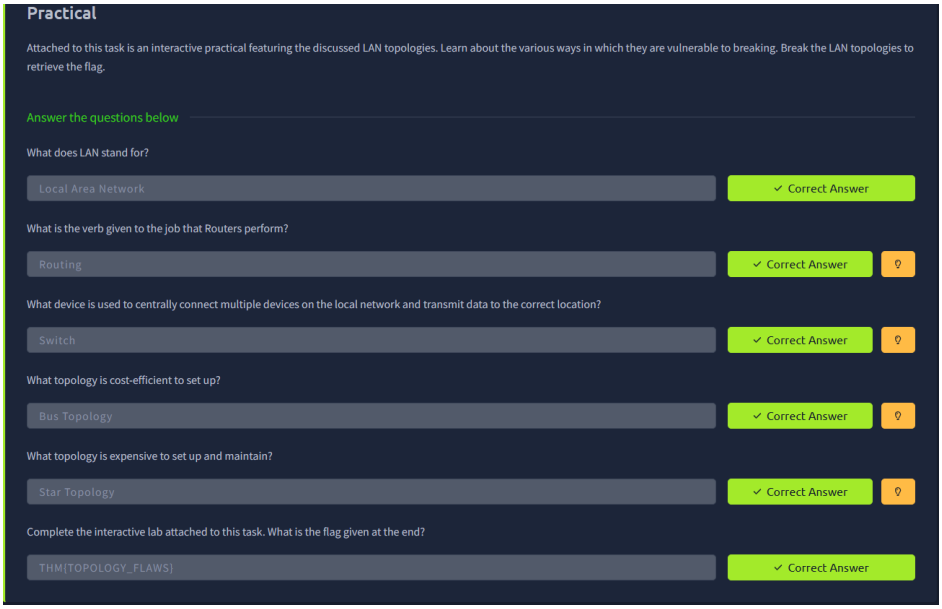


Ring topology

Devices form a logical or physical ring. The impact of a failure depends on the ring technology and its redundancy mechanisms.



A switch centrally connects devices in a typical switched LAN and forwards Ethernet frames using MAC addresses. A router forwards IP packets between networks.



Step 2 - Understand subnetting

Subnetting divides a larger IP network into smaller networks.

An IPv4 address contains 32 bits. A subnet mask identifies the network portion and host portion of the address.

Example:

IP address: 192.168.1.25

Subnet mask: 255.255.255.0

CIDR: /24

For this example:

- Network address: 192.168.1.0
- Example host address: 192.168.1.25
- Broadcast address: 192.168.1.255
- Typical usable host range: 192.168.1.1–192.168.1.254



Step 4 - Learn ARP

ARP stands for Address Resolution Protocol.

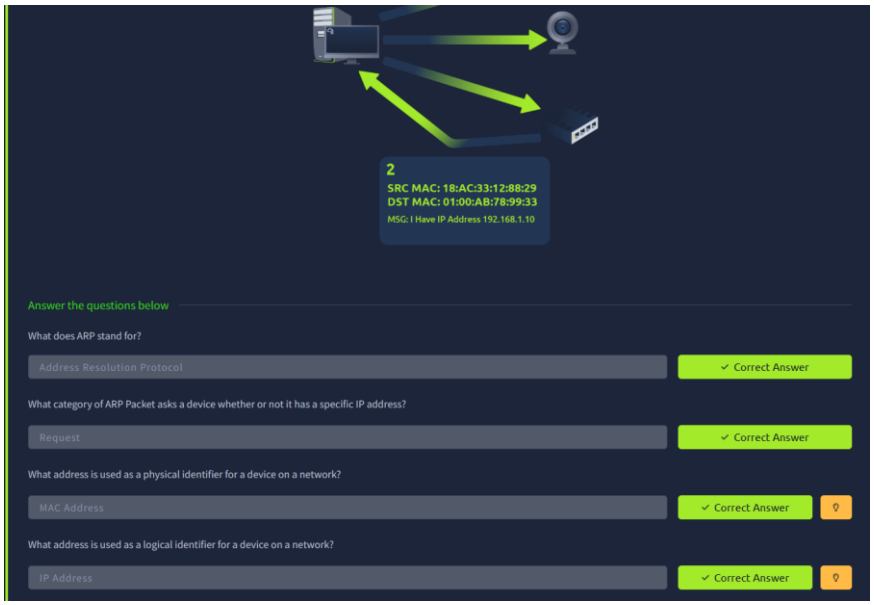
On a typical IPv4 Ethernet LAN, ARP helps a device discover the MAC address associated with a local IPv4 address.

The process works as follows:

1. **ARP Request**: The sender broadcasts a request asking which device owns the target IPv4 address.
2. **ARP Reply**: The device with the requested address normally replies with its MAC address.
3. **ARP Cache**: The sender can cache the IP-to-MAC mapping for subsequent communication.

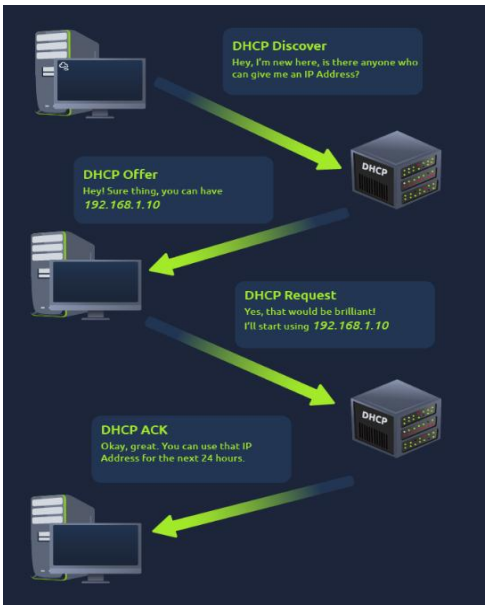
Useful Linux command: ip neigh

This displays the neighbor table, which includes learned local address mappings.



Step 5 - Learn DHCP

DHCP stands for Dynamic Host Configuration Protocol. It allows devices to obtain network configuration automatically.



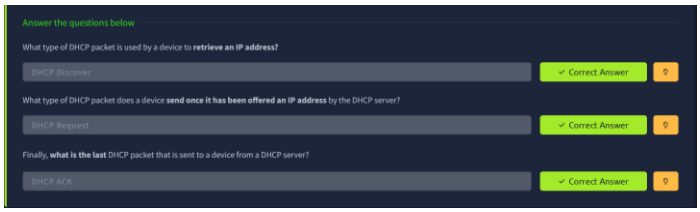
The common four-step DHCP exchange is known as **DORA**:

D -> Discover: The client searches for DHCP servers.

O -> Offer: A DHCP server offers configuration.

R -> Request: The client requests the offered configuration.

A -> Acknowledgment (ACK): The server confirms the configuration lease.



DHCP can supply an IP address, subnet mask, default gateway, and DNS server settings.

Step 6 - Complete the topology lab

1. Launch the interactive topology exercise in the room.
2. Inspect how the devices are connected.
3. Identify the central networking device.
4. Follow the lab instructions to complete the topology challenge.
5. Record the flag displayed at the end, if the current exercise asks for one.

What you should learn: A LAN relies on switches and link-layer addressing for local delivery, IP addressing and subnetting for logical networks, ARP for IPv4-to-MAC resolution, and DHCP for automatic configuration.